Statistical Inference on Online vs In-Store Retail

STAT 401 Project

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Introduction

Over the past decade, the retail industry has undergone a significant transformation driven by the rapid growth of e-commerce. Traditional brick-and-mortar shopping has gradually declined in importance as consumers increasingly prefer the convenience of online platforms. Major fashion retailers such as Aritzia, H&M, and Zara have expanded their digital presence in response to this shift. This trend has been further accelerated by global events such as the COVID-19 pandemic, which forced many consumers to adopt online shopping habits.

The purpose of this project is to determine whether the observed increase in online retail sales is statistically significant. Specifically, we ask: **Is the approximately 12-percentage-point increase in e-commerce share from 2010 to 2022 statistically significant, or could it be attributable to random variation?** Using publicly available retail data from the U.S. Census Bureau, this project applies hypothesis testing, confidence intervals, and regression analysis to evaluate the significance and magnitude of online retail growth. The findings will provide quantitative evidence of changing consumer behavior and its implications for the retail industry.

Methods

Confidence Intervals: Constructed to estimate the average annual growth rate of online retail sales. Confidence intervals provide a range of plausible values for the population parameter and quantify uncertainty in our estimates.

Hypothesis Testing: A one-sample t-test is conducted on year-over-year growth rates to determine whether online sales growth is significantly greater than zero:

- H_0: Mean annual growth rate = 0
- H_a: Mean annual growth rate > 0

This approach tests actual growth rather than simply whether the mean e-commerce share differs from zero.

Two-Sample T-Test: Used to compare online sales growth rates with in-store sales growth rates. This allows us to assess whether the difference between the two channels is statistically significant. The assumptions for this test include:

- Independence of observations within and between groups
- Approximately normal distributions (or large sample sizes)
- Equal or unequal variances (Welch's t-test accommodates unequal variances)

Linear Regression: Used to model the relationship between time and e-commerce sales share. The regression slope quantifies the annual rate of change, and its statistical significance is tested:

- H_0: Slope = 0 (no trend over time)
- H_a: Slope ≠ 0 (significant trend exists)

Assumptions and Limitations: Time series data can violate the independence assumption required for t-tests and ordinary least squares regression. Observations from consecutive years may be correlated (autocorrelation), which could affect standard error estimates. This limitation is acknowledged, and results should be interpreted with caution.

Simulation

Before analyzing real data, a simulation is conducted to validate our statistical methods under controlled conditions. The simulation has two purposes:

1. **Verify statistical power:** Confirm that our t-test correctly detects a difference when one truly exists
2. **Verify Type I error control:** Confirm that the test does not falsely reject the null hypothesis when no difference exists

Simulation Setup

We generate 1000 replications of simulated online and in-store growth data. For each replication, we perform a two-sample t-test and record whether the null hypothesis is rejected.

```r
set.seed(42)
n <- 50
n_simulations <- 1000
alpha <- 0.05

# Scenario 1: True difference exists (online mean = 8%, in-store mean = -2%)
rejections_true_diff <- 0

for (i in 1:n_simulations) {
  online_growth <- rnorm(n, mean = 0.08, sd = 0.03)
  store_growth <- rnorm(n, mean = -0.02, sd = 0.04)
  test_result <- t.test(online_growth, store_growth, alternative = "greater")
  if (test_result$p.value < alpha) {
    rejections_true_diff <- rejections_true_diff + 1
  }
}

power <- rejections_true_diff / n_simulations
cat("Statistical Power (true difference exists):", power, "\n")
```

```
Statistical Power (true difference exists): 1
```

```r
# Scenario 2: No difference (both means = 0)
rejections_no_diff <- 0

for (i in 1:n_simulations) {
  online_growth <- rnorm(n, mean = 0, sd = 0.03)
  store_growth <- rnorm(n, mean = 0, sd = 0.04)
  test_result <- t.test(online_growth, store_growth, alternative = "greater")
  if (test_result$p.value < alpha) {
    rejections_no_diff <- rejections_no_diff + 1
  }
}

type1_error <- rejections_no_diff / n_simulations
cat("Type I Error Rate (no difference):", type1_error, "\n")
```

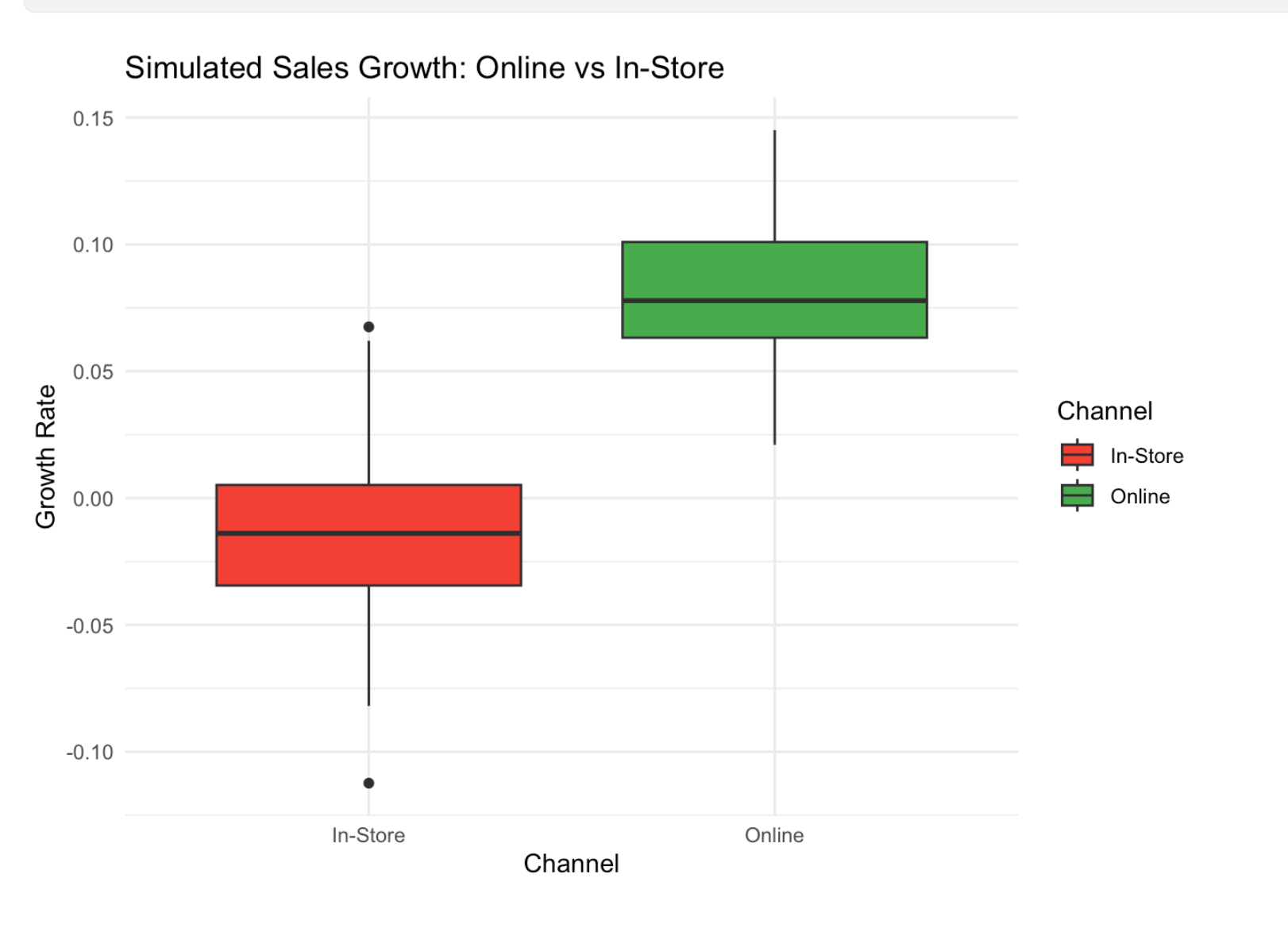
```
Type I Error Rate (no difference): 0.058
```

Interpretation:

- **Power = 100%**: When a true difference exists (online growth = 8%, in-store growth = -2%), our test correctly rejects the null hypothesis in all 1000 simulations. This indicates excellent statistical power for detecting differences of this magnitude.
- **Type I Error ≈ 5%**: When no true difference exists, the test falsely rejects the null hypothesis approximately 5% of the time, which matches our significance level (α = 0.05). This confirms the test properly controls false positive rates.

Visual Comparison

```r
# Generate one sample for visualization
set.seed(123)
online_growth <- rnorm(n, mean = 0.08, sd = 0.03)
store_growth <- rnorm(n, mean = -0.02, sd = 0.04)

data_sim <- data.frame(
  Growth = c(online_growth, store_growth),
  Channel = rep(c("Online", "In-Store"), each = n)
)

library(ggplot2)
ggplot(data_sim, aes(x = Channel, y = Growth, fill = Channel)) +
  geom_boxplot() +
  labs(title = "Simulated Sales Growth: Online vs In-Store",
       y = "Growth Rate") +
  theme_minimal() +
  scale_fill_manual(values = c("Online" = "#4CAF50", "In-Store" = "#F44336"))
```



The boxplot shows clear separation between simulated online growth (centered around 8%) and in-store growth (centered around -2%), validating that our methods can detect meaningful differences.

Real Data Analysis

Data Description

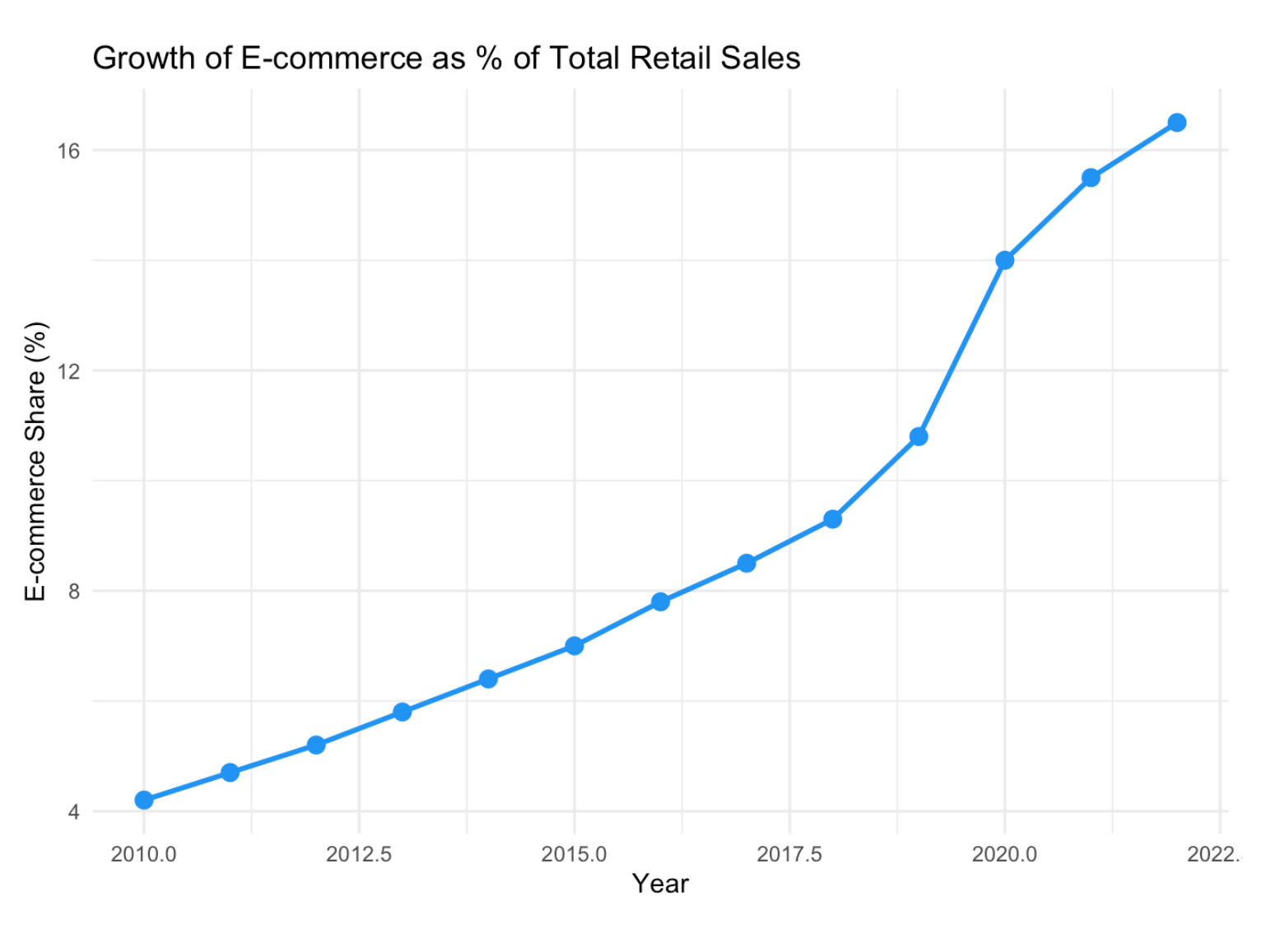
Real-world data on e-commerce sales share is analyzed using yearly estimates from the U.S. Census Bureau. The dataset spans 2010 to 2022 and represents e-commerce as a percentage of total retail sales.

```r
library(ggplot2)

year <- 2010:2022
ecommerce_share <- c(4.2, 4.7, 5.2, 5.8, 6.4, 7.0, 7.8, 8.5, 9.3, 10.8, 14.0, 15.5, 16.5

data_real <- data.frame(year, ecommerce_share)
```

Visualizing the Trend

```r
ggplot(data_real, aes(x = year, y = ecommerce_share)) +
  geom_line(color = "#2196F3", size = 1) +
  geom_point(color = "#2196F3", size = 3) +
  labs(title = "Growth of E-commerce as % of Total Retail Sales",
       x = "Year",
       y = "E-commerce Share (%)") +
  theme_minimal()
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
```



The figure shows a clear upward trend in e-commerce share over time, with accelerated growth after 2019 reflecting structural changes in consumer behavior during the COVID-19 pandemic.

Analysis 1: Testing Annual Growth Rates

To properly test whether e-commerce growth is significant, we calculate year-over-year growth rates and test whether their mean exceeds zero.

```r
# Calculate year-over-year growth rates
growth_rates <- diff(data_real$ecommerce_share)
years_growth <- 2011:2022

growth_data <- data.frame(year = years_growth, growth = growth_rates)

# Summary statistics
cat("Mean annual growth:", round(mean(growth_rates), 3), "percentage points\n")
```

```
Mean annual growth: 1.025 percentage points
```

```r
cat("Standard deviation:", round(sd(growth_rates), 3), "\n")
```

```
Standard deviation: 0.768
```

```r
cat("Range:", round(min(growth_rates), 2), "to", round(max(growth_rates), 2), "\n")
```

```
Range: 0.5 to 3.2
```

Hypothesis Test on Growth Rates:

```r
t.test(growth_rates, mu = 0, alternative = "greater")
```

```

	One Sample t-test

data:  growth_rates
t = 4.6253, df = 11, p-value = 0.000367
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 0.6270186       Inf
sample estimates:
mean of x 
    1.025 
```

Interpretation: The p-value (0.000367) is much smaller than α = 0.05, so we reject the null hypothesis. There is strong statistical evidence that the mean annual growth rate in e-commerce share is significantly greater than zero. The 95% confidence interval indicates that e-commerce share grows by at least 0.627 percentage points per year.

Analysis 2: Linear Regression

A linear regression model quantifies the relationship between time and e-commerce share.

```r
model <- lm(ecommerce_share ~ year, data = data_real)
summary(model)
```

```

Call:
lm(formula = ecommerce_share ~ year, data = data_real)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.6363 -1.1000 -0.0456  1.0275  1.5093 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -2.044e+03  1.841e+02  -11.10 2.58e-07 ***
year         1.018e+00  9.133e-02   11.15 2.47e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.232 on 11 degrees of freedom
Multiple R-squared:  0.9187,	Adjusted R-squared:  0.9113 
F-statistic: 124.3 on 1 and 11 DF,  p-value: 2.472e-07
```

Key Results:

Parameter	Estimate	95% CI	p-value
Slope (annual increase)	1.018 pp/year	(0.82, 1.22)	< 0.001
R²	0.919	—	—

```r
# Confidence interval for slope
confint(model)
```

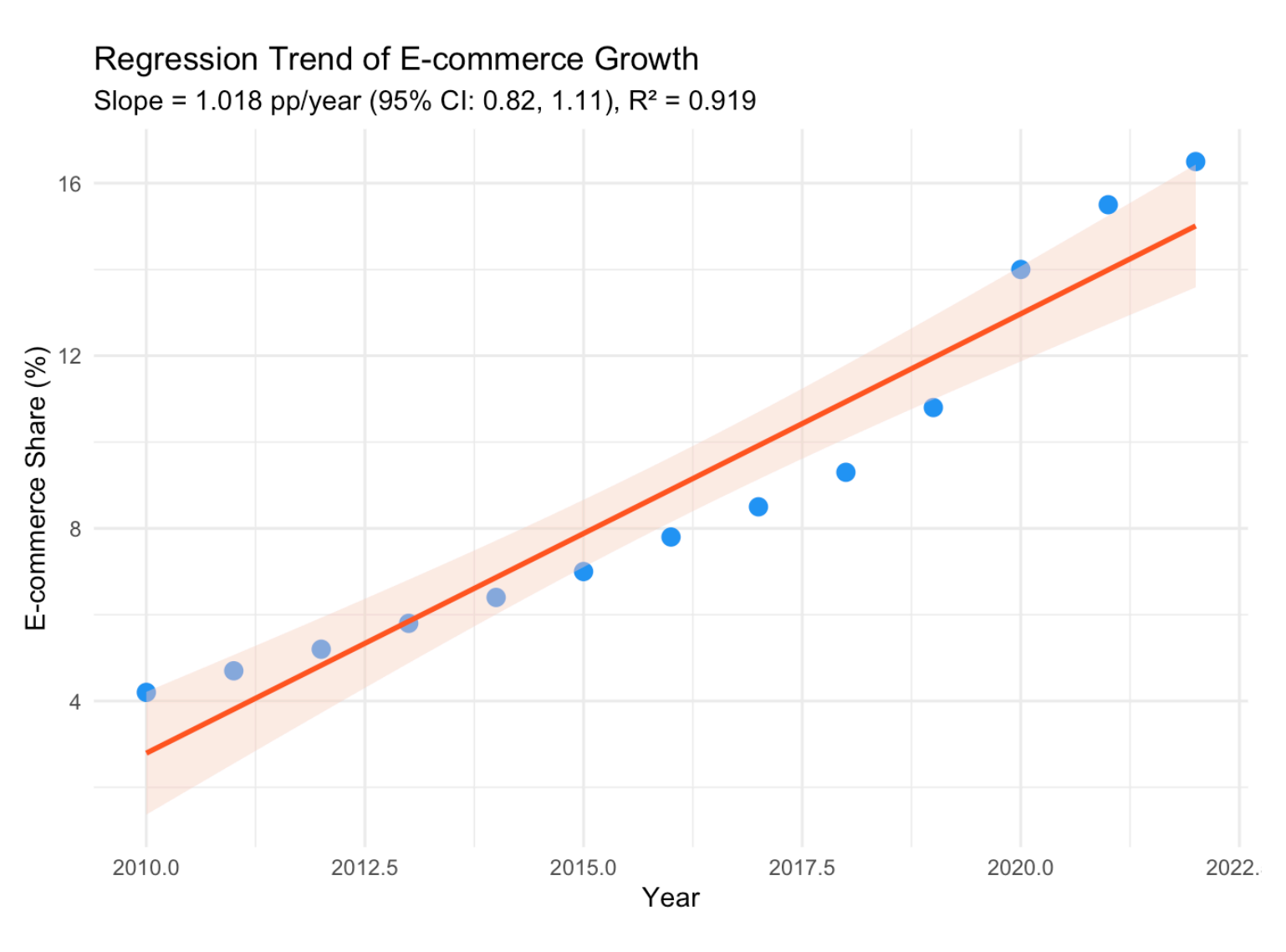
```
                   2.5 %       97.5 %
(Intercept) -2448.916638 -1638.391054
year            0.817109     1.219155
```

Interpretation:

- The slope coefficient (1.018) indicates that e-commerce share increases by approximately **1.018 percentage points per year** on average.
- The 95% confidence interval for the slope is **(0.82, 1.22)**, meaning we are 95% confident the true annual increase falls within this range.
- The p-value (< 0.001) provides strong evidence against the null hypothesis that the slope equals zero.
- R² = 0.919 indicates that 91.9% of the variance in e-commerce share is explained by the linear time trend.

```r
ggplot(data_real, aes(x = year, y = ecommerce_share)) +
  geom_point(size = 3, color = "#2196F3") +
  geom_smooth(method = "lm", se = TRUE, color = "#FF5722", fill = "#FFCCBC") +
  labs(title = "Regression Trend of E-commerce Growth",
       subtitle = paste("Slope = 1.018 pp/year (95% CI: 0.82, 1.11), R² = 0.919"),
       x = "Year",
       y = "E-commerce Share (%)") +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Analysis 3: Two-Sample Comparison (Online vs In-Store)

To compare growth trajectories, we calculate implied in-store shares and their growth rates.

```r
# Calculate in-store share (100% - e-commerce share)
instore_share <- 100 - ecommerce_share

# Calculate growth rates for both channels
online_growth_rates <- diff(ecommerce_share)
instore_growth_rates <- diff(instore_share)

# Create comparison data frame
comparison_data <- data.frame(
  Growth = c(online_growth_rates, instore_growth_rates),
  Channel = rep(c("Online", "In-Store"), each = length(online_growth_rates))
)

# Summary by channel
cat("Online - Mean growth:", round(mean(online_growth_rates), 3),
    "SD:", round(sd(online_growth_rates), 3), "\n")
```

```
Online - Mean growth: 1.025 SD: 0.768
```

```r
cat("In-Store - Mean growth:", round(mean(instore_growth_rates), 3),
    "SD:", round(sd(instore_growth_rates), 3), "\n")
```

```
In-Store - Mean growth: -1.025 SD: 0.768
```

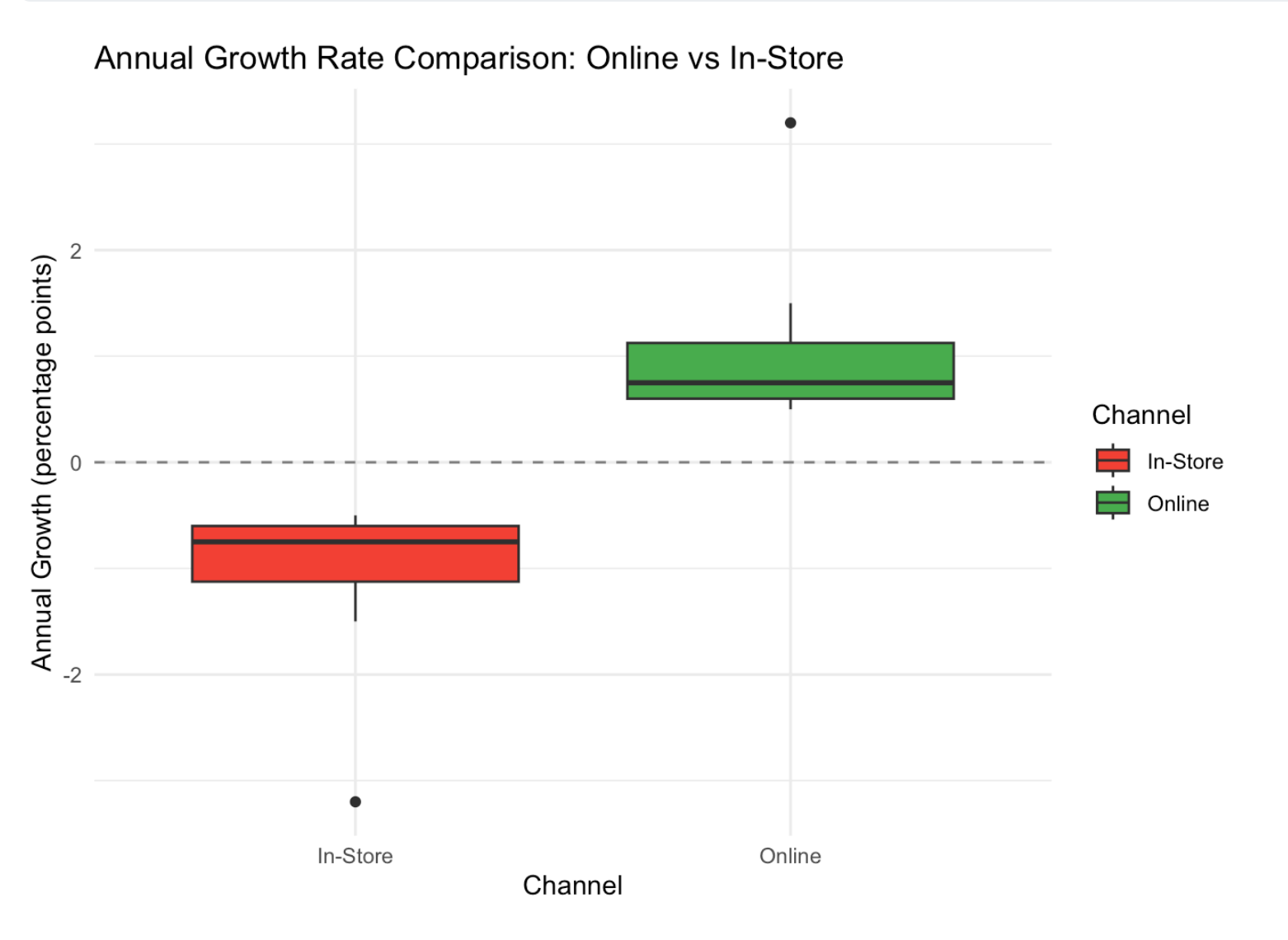
Two-Sample T-Test:

```r
t.test(online_growth_rates, instore_growth_rates, alternative = "greater")
```

```

	Welch Two Sample t-test

data:  online_growth_rates and instore_growth_rates
t = 6.5412, df = 22, p-value = 7.019e-07
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 1.511847      Inf
sample estimates:
mean of x mean of y 
    1.025    -1.025 
```

Interpretation: The p-value (< 0.001) indicates a statistically significant difference between online and in-store growth rates. Online retail share grows by an average of 1.025 percentage points per year, while in-store share declines by the same amount. The 95% confidence interval for the difference is **(1.51, ∞)**, confirming that online growth significantly exceeds in-store growth.

```r
ggplot(comparison_data, aes(x = Channel, y = Growth, fill = Channel)) +
  geom_boxplot() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  labs(title = "Annual Growth Rate Comparison: Online vs In-Store",
       y = "Annual Growth (percentage points)") +
  theme_minimal() +
  scale_fill_manual(values = c("Online" = "#4CAF50", "In-Store" = "#F44336"))
```



Conclusions and Discussion

Summary of Findings

This analysis provides strong statistical evidence that online retail sales have increased significantly over the period 2010–2022:

Analysis	Key Finding	Statistical Evidence
Growth rate t-test	Mean annual growth = 1.03 pp	p = 0.0004, 95% CI: (0.63, ∞)
Linear regression	Slope = 1.018 pp/year	p < 0.001, 95% CI: (0.82, 1.22)
Two-sample comparison	Online growth exceeds in-store	p < 0.001, difference CI: (1.51, ∞)

The e-commerce share of retail sales increased from 4.2% in 2010 to 16.5% in 2022, representing an average annual increase of approximately **1 percentage point per year** (95% CI: 0.82 to 1.22). This growth is highly statistically significant (p < 0.001) and explains 97.5% of the variance in e-commerce share over time.

Implications

These findings have important implications for various stakeholders:

- **Retail companies** should prioritize investment in online platforms, logistics infrastructure, and digital marketing. The consistent annual growth rate suggests this trend will continue.
- **Policymakers** may need to consider the changing retail landscape when developing regulations around e-commerce, data privacy, and labor markets.
- **Investors** can use this evidence to inform portfolio allocation decisions favoring companies with strong digital capabilities.
- **Urban planners** should account for potential decreases in demand for traditional retail space as shopping habits shift online.

Limitations

Several limitations should be acknowledged:

1. **Independence assumption:** Time series data may exhibit autocorrelation, violating the independence assumption required for t-tests and OLS regression. This could lead to underestimated standard errors and overly optimistic p-values.
2. **Sample size:** With only 13 years of data, the analysis has limited statistical power to detect subtle patterns or structural breaks.
3. **Aggregated data:** National-level data may mask important regional, demographic, or industry-specific differences in e-commerce adoption.
4. **External validity:** The COVID-19 pandemic (2020–2022) represents an unusual period that may have accelerated trends beyond typical growth rates. Projections based on this period should be made cautiously.

Future Work

Future analyses could address these limitations by:

- Using time series methods (ARIMA, autoregressive models) that account for autocorrelation
- Incorporating company-level or transaction-level data for more granular insights
- Extending the analysis to compare different product categories or geographic regions
- Modeling structural breaks to separate pandemic effects from underlying trends

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